

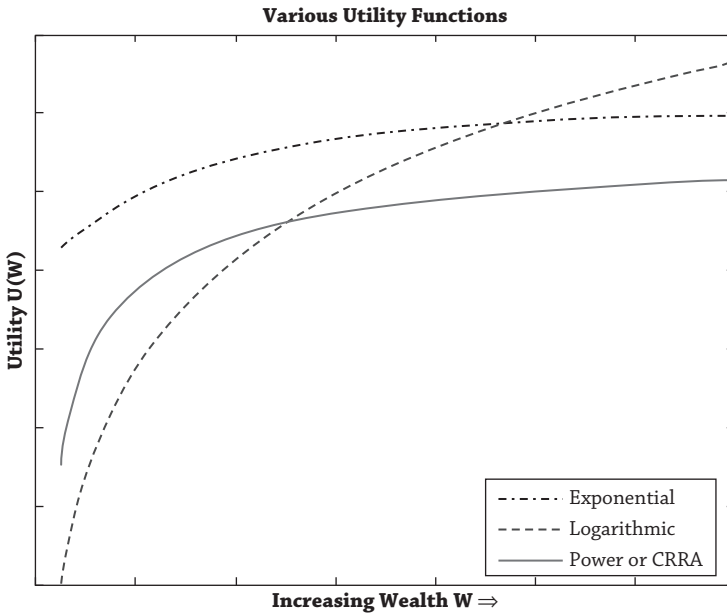


Figure 2.3

I have purposely left the units off the x - and y -axes. These utility functions are *ordinal*, so that the actual numerical value of utility has no meaning. The numbers merely denote ranks: we can replace the utility function U by $a + bU$ for any positive constants a and b without changing the preference ordering. (That is, utility orderings are unchanged by *affine transformations*.)

All of the utility functions in Figure 2.3 increase with wealth, which realistically reflects the fact that asset owners are generally greedy. Furthermore, utility functions are all *concave* over wealth. Concavity is a measure of how much investors value an extra \$1 of wealth. When investors are poor, we sit close to the y -axis in Figure 2.3, and the slopes of all the curves are very steep. When you only have \$1, you really value going from \$1 to \$2. When wealth is high, the utility curves flatten out: when you already have \$10 million, there is little to be gained from going from \$10,000,000 to \$10,000,001 in wealth. The utility curves exhibit *diminishing marginal utility* in wealth. This is an appealing property of concave utility functions.

The slope of the utility function (the changing marginal utility as wealth increases) is how we measure just how bad or good the investor feels. Utilities that decrease very fast as wealth decreases correspond to investors for whom losses really hurt. Asset owners who have utilities that are very flat as wealth increases don't care much for increasing their wealth when wealth is already very high. *Bad times are times of high marginal utility*. Bad times are when the utility curve is very steep: at these points the asset owner really, really wants another \$1 ("Brother can